LUMI MINING

DOCUMENT	LMW-SPEC-001
REVISION	A — for quotation
DATE	2026-07-16
UNITS	mm (unless noted)
DRAWINGS	GA-01 … GA-05

PARTS & SERVICES DIVISION

Mobile Workshop Container

Manufacturing design & specification package for fabrication and fit-out.
Base unit LMW-20S; derivatives LMW-20HC and LMW-40P.

CONFIDENTIAL — ISSUED TO THE FABRICATION PARTNER FOR QUOTATION AND BUILD. DO NOT REDISTRIBUTE.



LMW-20S · 20 FT ISO
RAL 7021 / RAL 1004

GA-00 · EXTERIOR GENERAL ARRANGEMENT — ISOMETRIC	SCALE NTS	DIMS MM	REV A

SECTION 1

Scope of supply

Fabricate and fit out a self-contained, field-deployable maintenance workshop inside an ISO shipping container, to the general arrangement drawings GA-01…GA-05 and the systems specifications in Sections 3–7. The base unit LMW-20S (20 ft standard, one-trip container) is the subject of this package; LMW-20HC and LMW-40P share the shell and services architecture with the additions listed below and will be released as separate build orders.

CONFIG	NAME	SHELL	DEPLOYMENT	ADDITIONS OVER BASE
LMW-20S	Field Tech	20 ft Standard	Underground · single technician	Base unit — this document
LMW-20HC	Underground Pro	20 ft High Cube	Underground · two-tech hub	20 t press, canopy hose crimper, RFID parts cabinet, gas detection
LMW-40P	Site Workshop	40 ft High Cube, partitioned	Surface / open pit · multi-tech	Crib room ≥ 3 m² (kitchenette, A/C), dual zone, vehicle-hoist prep

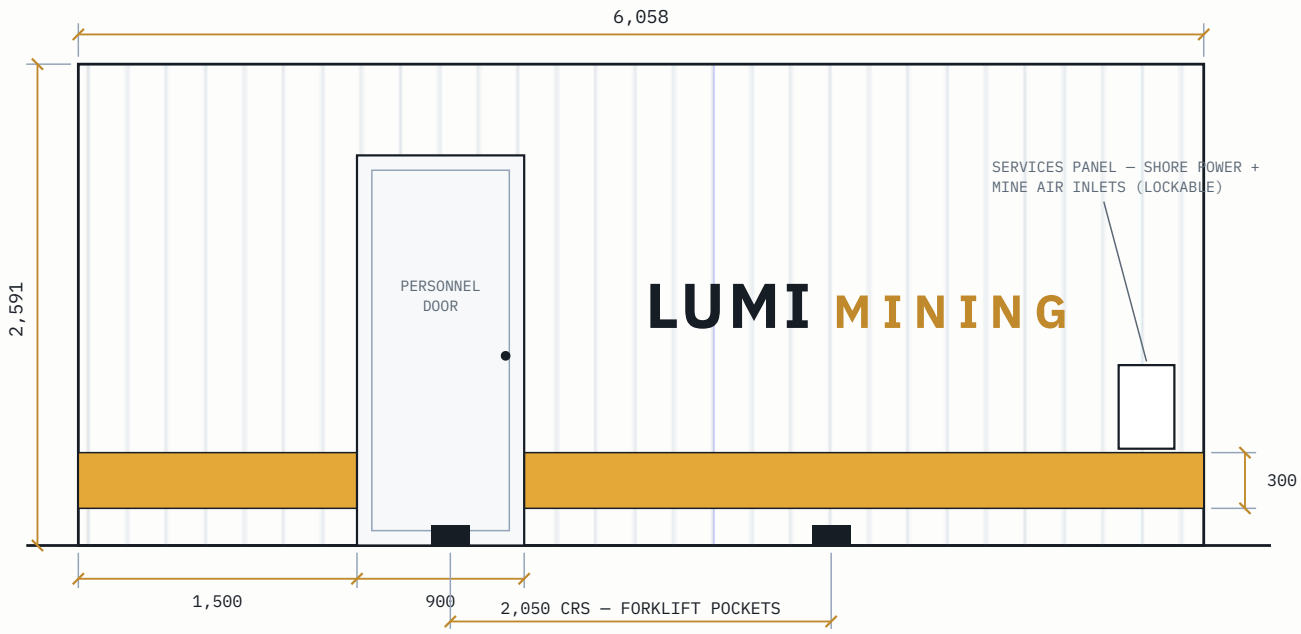
SHELL	EXTERNAL L × W × H	INTERNAL L × W × H	BEST FOR
20 ft Standard	6,058 × 2,438 × 2,591	5,900 × 2,350 × 2,390	Underground, tight access
20 ft High Cube	6,058 × 2,438 × 2,896	5,900 × 2,350 × 2,700	Ergonomics, headroom permitting
40 ft High Cube	12,192 × 2,438 × 2,896	12,032 × 2,350 × 2,700	Premium fit-out, dual zone

Underground transit requires tunnel profile ≥ 2.6 m H × 2.6 m W — confirm site clearance before substituting a high-cube shell. Donor container: one-trip, CSC-plated, ISO 668 / ISO 1161 compliant, surveyed free of dents and floor delamination.

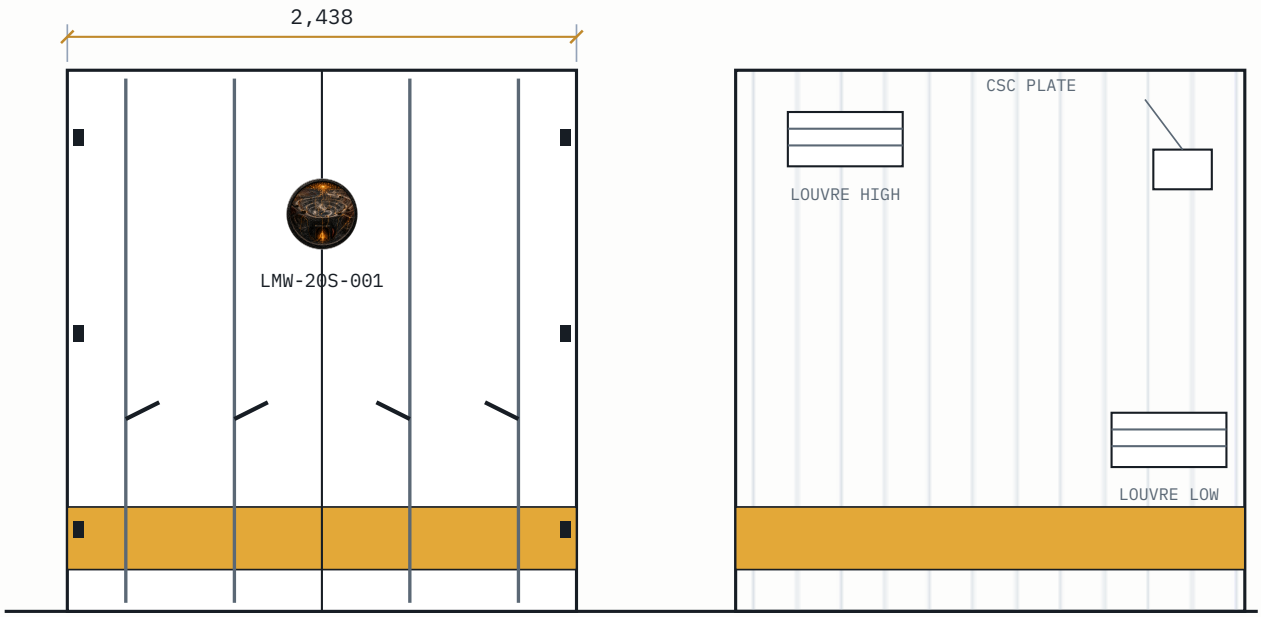
SECTION 2

General arrangement drawings

All dimensions in millimetres from the external corner castings unless noted. Fabrication tolerance ±3 mm on openings, ±10 mm on fit-out positions. Where a drawing conflicts with a table, the table governs; flag discrepancies before cutting.



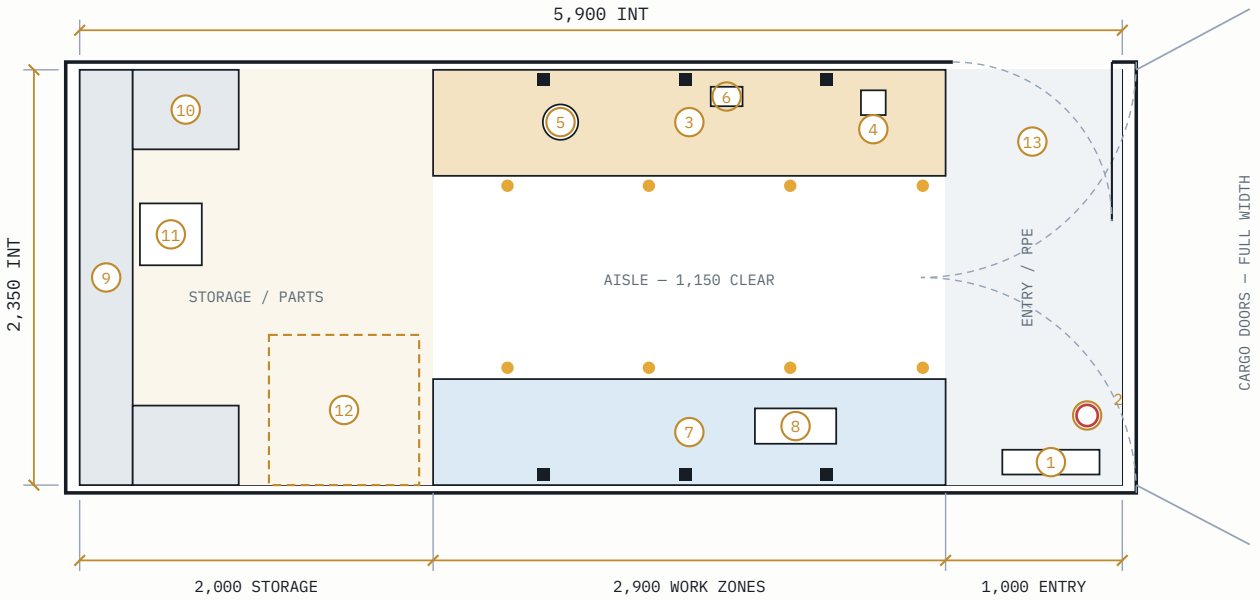
GA-01 · SIDE ELEVATION — KERB SIDE (PERSONNEL DOOR WALL) | SCALE 1:10 @ VIEWBOX | DIMS MM



A — DOOR END · 3-POINT LOCKING, 180° SWING

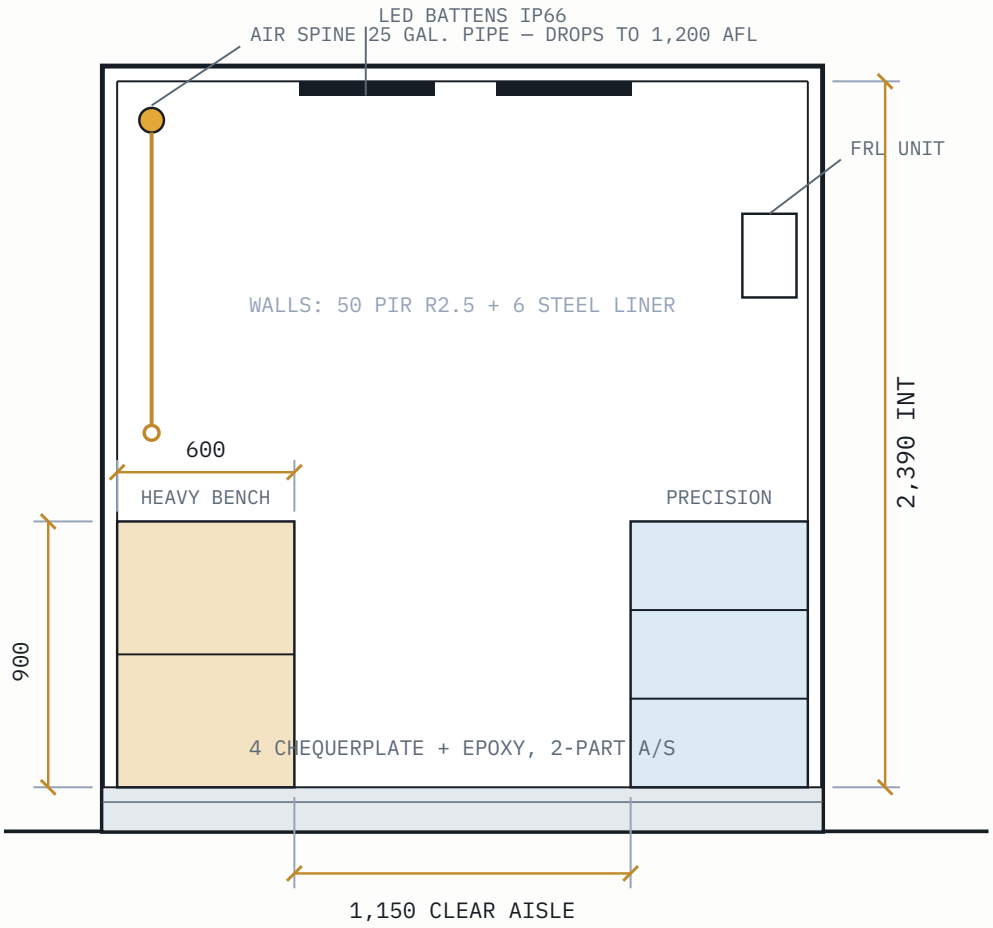
B — REAR END · IP54 VENT LOUVRES HIGH + LOW

GA-02 · END ELEVATIONS — DOOR END (A) AND REAR END (B) | DIMS MM



GA-03 · FLOOR PLAN — INTERNAL FIT-OUT, LMW-20S | DIMS MM

1 — PPE RACK + FIRST AID
2 — FIRE EXTINGUISHER (ABE)
3 — HEAVY BENCH 2,900×600
4 — BENCH VICE 150
5 — PILLAR DRILL 16
6 — BENCH GRINDER
7 — PRECISION BENCH 2,900×600
8 — DIAGNOSTIC STATION
9 — SHELVING, 300 DEEP
10 — LOCKABLE CAGES ×2
11 — 20 T PRESS
12 — BUNDED FLUID ZONE
13 — WHITEBOARD / PTW
● AIR DROP ×8
■ 240 V GPO ×6



GA-04 · CROSS-SECTION — LOOKING TOWARD DOOR END

DIMS MM · AFL = ABOVE FINISHED FLOOR

SECTION 3

Structure, openings & climate

OPENING	LOCATION	SPECIFICATION
Double cargo doors	Door end	Retain OEM full-width doors; 180° swing, 3-point locking, service gaskets, add padlock hasp + door retainers
Personnel door	Kerb-side wall, per GA-01	900 × 2,100 clear; steel frame welded to framed cut-out; anti-panic bar inside, keyed lever outside; self-closing; opens outward
Vent louvres ×2	Rear end, high + low	550 × 260 IP54 louvres, vermin mesh, framed and sealed
Exhaust fan	Rear end, high	300 axial fan behind high louvre, switched at entry; auto-run on gas-detect option
Services panel	Kerb-side wall, door end	Lockable recessed box: shore-power inlet + mine-air inlet + E-stop, per GA-01 and Sections 4–5

FLOORING BUILD-UP

- OEM corrugated steel base + marine ply substrate
- 4 mm chequerplate steel inlay, fully welded seams
- 2-part epoxy finish — anti-slip, oil and chemical resistant
- Bunded fluid zone: 300 upstand, drain valve to collection point (GA-03 key 12)
- Load rating ≥ 3 t/m² general, ≥ 5 t/m² at forklift entry

INSULATION & CLIMATE

- Walls + ceiling: 50 mm PIR board, R2.5 min, foil vapour barrier at ceiling
- Liner: 6 mm steel (or 9 mm OSB) over insulation, impact resistant
- A/C: 18,000 BTU split system, reverse cycle; condenser on rear end wall, guarded
- Target ≤ 60% RH for electronics longevity

SECTION 4

Electrical system

ITEM	SPECIFICATION	QTY
Shore power inlet	415 V 3-phase 63 A industrial appliance inlet, in exterior lockable services panel	1
Distribution board	RCD-protected consumer unit, galvanized steel enclosure, circuit schedule laminated inside door	1
GPO 240 V 13 A double	IP53 double outlets at bench back-boards per GA-03; surge-protected strip at precision bench	6
Industrial socket	415 V 16 A 5-pin, adjacent to heavy bench for press / welder feed	1
Lighting	LED vapour-proof battens IP66, 2 rows × 2, ≥ 500 lux at bench height; IP66 2-way switches at both doors	4
E-stop isolation	Exterior E-stop in services panel, lockout/tagout capable, isolates all circuits except lighting	1
Options (per order)	110 V low-voltage tool circuit · UPS for diagnostics · roof solar prep (conduit + mounts only)	—

All wiring in galvanized conduit or cable tray above liner; no exposed flex. Test and tag to local code; supply certificate of electrical compliance with the unit.

SECTION 5

Compressed air system

Supply	Mine-air inlet in exterior services panel — ¾" BSP female + quick-release coupling, isolation ball valve inside
Distribution	25 mm galvanized pipe spine at ceiling (GA-04), drops to 1,200 AFL at each service point
Service points	8× ¼" BSP quick-release outlets, 4 per wall per GA-03; one swivel hose reel (15 m) on overhead rail
Regulation	In-line FRL (filter + regulator + lubricator) at inlet; regulated to 90 PSI / 6.2 bar at tool outlets; gauge visible from aisle
Option	Onboard compressor 15–22 CFM on anti-vibration mounts, rear storage zone, ducted intake — for sites without mine air

SECTION 6

Interior fit-out — bill of equipment

Positions per GA-03 keys. Items marked ⬡ are supplied by Lumi Mining and fitted by the fabricator; all other items are fabricator supply-and-fit.

KEY	ITEM	SPECIFICATION	QTY
3	Heavy workbench	2,900 × 600 × 900 H; 50 hardwood top on steel frame (or 3 steel plate); ≥ 1,000 kg rating; bolted to wall + floor; cable tray under	1
7	Precision workbench	2,900 × 600 × 900 H; 3× lockable A4-DIN drawer units; anti-static mat surface; surge-protected power strip	1
4	Bench vice	150 jaw, swivel base, bolted through bench top	1
5	Pillar drill	16 chuck, variable speed, bench-mounted with guard	1
6	Bench grinder mount	200 grinder on bolt-down pedestal plate; spark guard; fire blanket within 1 m	1
11	Hydraulic press	20 t floor-standing shop press, anchored; guarding to local code	1
9	Shelving bays	Floor-to-ceiling, 4 bays, 300 deep, ≥ 200 kg/shelf; Eurobin rail rows above precision bench	4
10	Lockable steel cages	Welded mesh, keyed alike, for consumables and high-value parts	2
—	Tool shadow board	Perforated steel panel above heavy bench, full length, powder-coated RAL 7021	1
—	Parts trolley	6–8 drawer, castored, lockable; parks under heavy bench end	1
—	Torque station	Wall rack for 3 torque wrenches: 25–150 / 100–600 / 300–1,500 Nm ⬡	1
8	Diagnostic station	Laptop dock + dual monitor arms; diagnostic PC, CAN logger, test kit ⬡; Wi-Fi/LTE router mount + antenna gland	1
12	Fluid management	Oil drain trolley 50 L; dispensing guns (engine, hydraulic, grease); drip tray 110% of largest container; 200 L waste-oil barrel positioned exterior	1 set
1·2	Safety set	2× extinguishers (ABE at entry, CO_2 at electronics); Class C first aid; 16 L eyewash; MSDS holder at entry; fire blanket at grinder; statutory signage kit	1 set

SECTION 7

Digital layer — factory provisions

Lumi installs and commissions the digital equipment (CMMS terminal, RFID cabinet, gas detection, sensors) after delivery. The fabricator provides the physical provisions below so installation is plug-in.

- 20 mm conduit runs: services panel → DB → both bench back-boards → entry door frame
- Router mount plate + external antenna gland (roof, sealed) for LTE/Wi-Fi bridge
- VESA mount at entry for permit-to-work / status display
- Badge-reader cut-out + conduit at personnel door
- Fused spur + mount for gas detection head (CO + NO_2) with relay wired to exhaust fan
- RFID cabinet position: storage zone, 600 W × 450 D footprint, dedicated GPO
- Spare 4-way GPO behind diagnostic station for CMMS hardware

SECTION 8

Livery & branding



GA-05 · LIVERY — BOTH LONG WALLS IDENTICAL; UNIT ID ON BOTH DOOR LEAVES AND REAR END | DIMS MM

RAL 7021 Black Grey Body — all exterior faces + roof	**RAL 1004 Golden Yellow** Livery band, door edges, handles	**RAL 9002 Grey White** Interior liner + ceiling	**RAL 9005 Jet Black** Underframe, corner castings, pockets

- Paint system: abrasive blast SA 2.5 → zinc-rich epoxy primer → polyurethane topcoat, 200 µm min DFT; marine-grade for coastal sites.
- Decals: cast vinyl, laminated, reflective class 2. Wordmark decal reversed (white/gold) for dark body — artwork supplied by Lumi as vector files; do not recreate, rescale disproportionately, or recolor.
- Hex icon mark 340 × 340 on each cargo-door leaf with unit ID beneath, per GA-02. Clearspace around all marks ≥ 200 on the shell.
- Interior accent: shadow board and drawer fronts RAL 7021; safety zone floor markings golden yellow, 80 stripe.

SECTION 9

Transport, handling & marking

- 8× ISO corner castings retained, undamaged, twist-lock compatible
- Forklift pockets 2,050 crs per GA-01, min section 205 × 120; reinforce cut edges
- Lifting: 4-point chain sling via top castings, certified to CSC / ISO 3874
- 8× lashing D-rings, 8 t SWL each, on floor perimeter
- Max gross weight fully equipped ≈ 8,000 kg — record actual tare + payload on plate
- CSC plate revalidated after conversion; data plate with unit ID, tare, max gross riveted at door end
- Modes: road flatbed / lowboy, mine service vehicle, LHD carry frame
- Loose-shipped items (trolley, drain trolley, barrels) restrained for transit

SECTION 10

Compliance, testing & acceptance

TEST / CERTIFICATE	ACCEPTANCE CRITERION
Structure	ISO 668 / ISO 1161 castings intact; CSC plate revalidated; weld inspection report on all cut-outs
Weather tightness	Hose test, all openings and glands — zero ingress
Electrical	Certificate of compliance; RCD trip test; insulation resistance; lux survey ≥ 500 at bench
Compressed air	Pressure hold 8 bar / 30 min, ≤ 2% drop; 6.2 bar regulated at all outlets
Paint & livery	DFT readings ≥ 200 µm; colors match RAL references under daylight; decal positions per GA-05 ±10
Fit-out	Bench load test 1,000 kg; pull test on anchored equipment; photographic record against GA-03 keys
Documentation	As-built drawings, test certificates, paint batch records, O&M manual — delivered with unit

APPROVED FOR FABRICATION — LUMI MINING

Name / role Signature Date

ACCEPTED — FABRICATION PARTNER

Name / role Signature Date